

Structural Retrieval with Small Language Models Matches Amino-Acid Foundation-Model Performance on DGEb Enzyme Classification at Laptop Scale

Abhinav Erraguntla¹

¹Independent

Abstract

The DGEb protein classification suite [West-Roberts et al., 2024] evaluates amino-acid (AA) foundation models on functional tasks where sequence homology is wholly or partially removed between train and test. On Convergent Enzymes Classification (400 Enzyme Commission (EC) classes, sequence homology removed) the strongest published AA foundation model, ESM2-3B [Lin et al., 2023], reaches a weighted $F1 = 0.265$; on the easier homology-preserved EC Classification (128 classes) it reaches $F1 = 0.680$. This work shows that a confidence-gated combination of Foldseek structural retrieval [van Kempen et al., 2024] and a small protein-LM majority fallback reaches $F1 = 0.730$ on EC Classification (vs 0.680) and $F1 = 0.267$ on the harder, homology-removed Convergent Enzymes task (vs 0.265), with no task-specific training and on a CPU laptop with approximately one hour of free Colab GPU time for the two largest language-model embeddings. A bootstrap analysis places the DGEb baseline inside the 95% confidence interval of both results, so the pipeline matches a 3B-parameter foundation model at a fraction of the compute rather than beating it. A per-query analysis on Convergent Enzymes shows Foldseek’s per-hit alignment probability is the dominant separator of easy and hard queries (Mann–Whitney $p = 3 \times 10^{-5}$). Negative results are also reported: pocket-geometry features, discrete 3Di motif counting, and a Nucleotide Transformer v1 ensembling attempt all underperform. Code, embeddings, and a fully reproducible pipeline are public.¹

1 Introduction

Predicting an enzyme’s function from its primary sequence is a well-studied problem when the test enzyme has a homologous neighbour in the training set: the most-similar sequence’s annotation usually transfers [Gligorijević et al., 2021, Yu et al., 2023]. The *convergent* regime, in which enzymes catalyse the same reaction despite having independently evolved sequences and folds, is much harder, and only recently has a benchmark been published that specifically isolates it. The Diverse Genomic Embedding Benchmark (DGEb) [West-Roberts et al., 2024] includes a Convergent Enzymes Classification task with 2000 training and 400 test proteins drawn from 400 EC classes, with sequence homology removed between train and test within each class, and an EC Classification task with 512 training and 128 test proteins drawn from 128 EC classes, where train/test homology is permitted. The DGEb suite evaluates both AA and nucleotide-track foundation models, but the two classification tasks examined here are evaluated on AA models only; the strongest reported AA entry on each is ESM2-3B [Lin et al., 2023], at weighted $F1 = 0.265$ on Convergent Enzymes and $F1 = 0.680$ on EC.

¹<https://github.com/evrabhinav/convergent-enzymes-foldseek>

Recent analysis of convergent enzymes [Riziotis et al., 2025] describes three paradigms of functional convergence across a curated set of M-CSA cases: structural-and-mechanistic convergence, in which active sites align in 3D even when overall sequence and fold differ (14 of the 34 curated cases, $\sim 41\%$); mechanistic convergence without aligned catalytic residues; and reaction-only convergence. To the extent that the first paradigm dominates, structural-retrieval tools that operate on the local 3D environment, such as Foldseek’s 3Di alphabet [van Kempen et al., 2024], should recover much of what sequence-only models miss, without any task-specific training.

This paper makes the following empirical contributions.

- On EC Classification, a confidence-gated combination of Foldseek structural retrieval with a 4-LM majority fallback (ESM2-3B + ProstT5 + ESM2-150M + ESM2-35M) reaches $F1 = 0.730$ versus the ESM2-3B baseline of 0.680 (a +0.050 point gain; every τ -sweep configuration clears the baseline, though a bootstrap CI overlaps it, Section 4.1).
- On Convergent Enzymes, the same recipe with a 3-LM majority fallback (ESM2-3B + ProstT5 + ESM2-150M) reaches $F1 = 0.267$, matching the AA foundation-model baseline (0.265) with no training and a single AlphaFold structure per protein.
- To the author’s knowledge, this is the first reported Foldseek score on either of these DGEB classification tasks: $F1 = 0.238$ (top-1) on Convergent Enzymes and $F1 = 0.663$ on EC.
- A per-query analysis on Convergent Enzymes identifies what makes queries hard for the AA-track ensemble: Foldseek’s per-hit probability dominates as a separator ($p = 3 \times 10^{-5}$), with coding-sequence (CDS) length and protein length weaker, uncorrected effects.
- Documented negative results: amino-acid sequence features (424-D) score $F1 = 0.016$; hand-crafted global structural features (100-D) score 0.037; pocket-geometry features (fpocket, 81-D) score 0.019; discrete 3Di motif counting (linear k-mers and spatial pairs/triples) scores ≤ 0.073 ; Nucleotide Transformer v1 2.5B embeddings in this pipeline score 0.008 alone and reduce the ensemble F1 when added.

The contribution is to characterise the AA-foundation-model ceiling empirically on two DGEB tasks: with Foldseek’s structural retrieval as a base classifier and a small-LM majority fallback, that ceiling is reachable on a CPU laptop. The DGEB suite does evaluate nucleotide-track foundation models on other tasks (MIBIG Classification, EC DNA Classification, ribosomal and operonic tasks); the two AA-only classification tasks in this work are not part of that nucleotide-track benchmark.

2 Related Work

Function prediction from protein sequence. Methods such as DeepFRI [Gligorić et al., 2021] use graph convolutional networks on AlphaFold-derived contact maps, and CLEAN [Yu et al., 2023] contrasts EC-class prototypes against an ESM backbone. These methods score well on benchmarks where train and test share homology. The DGEB benchmark suite [West-Roberts et al., 2024] was introduced specifically to test what happens once the homology shortcut is removed.

Structural retrieval. Foldseek [van Kempen et al., 2024] encodes each residue’s local 3D environment as a token in a 20-letter alphabet (3Di) and reuses sequence-search machinery (Smith–Waterman, profile HMMs) on those tokens. It is several orders of magnitude faster than classical structural aligners such as TM-align [Zhang and Skolnick, 2005] or Dali [Holm and Sander, 1993], at comparable accuracy on the standard benchmarks reported by its authors. DGEB does not evaluate Foldseek.

Protein language models. ESM2 [Lin et al., 2023] is a family of masked-language-model transformers pretrained on UniRef sequences, with sizes from 8M to 15B parameters. ProstT5 [Heinzinger et al., 2024] is a bilingual T5 model trained jointly on amino-acid sequences and on Foldseek 3Di tokens, giving a single model that handles both modalities. The Nucleotide Transformer family [Dalla-Torre et al., 2025] and Evo [Nguyen et al., 2024] are nucleotide foundation models pretrained on genomic DNA; both are evaluated on DGEb’s nucleotide-track tasks (MIBIG Classification, EC DNA Classification, ribosomal and operonic tasks), but not on the two AA classification tasks examined in this work.

3 Method

3.1 Data

This study uses the DGEb `convergent_enzymes` dataset (2000 train, 400 test, 400 classes) and `ec_classification` dataset (512 train, 128 test, 128 classes). For each UniProt ID in either split, the AlphaFold predicted structure [Jumper et al., 2021, Varadi et al., 2024] is downloaded via the EBI API. Coverage is 98.3% on Convergent Enzymes (2360/2400 proteins) and 94.8% on EC Classification (607/640); the remaining proteins have no entry in AlphaFold DB.

3.2 Pipeline

The pipeline has three components.

Structural retrieval (Foldseek). A Foldseek database is built from the train set, and `easy-search` is run with default parameters (`-s 9.5, -e 10, --max-seqs 50`) and the standard 3Di scoring matrix. For each test query, the top-1 hit’s EC label is returned. Foldseek’s per-hit alignment probability `prob` serves as a confidence signal.

Amino-acid language-model fallbacks. Every protein is embedded with four LMs, each treated as a separate classifier head: ESM2-35M, ESM2-150M, and ESM2-3B [Lin et al., 2023], and ProstT5 in its AA mode [Heinzinger et al., 2024]. Embeddings are produced by mean-pooling the encoder’s last hidden state over residue positions, excluding the [CLS] and [EOS] special tokens. Sequences longer than the model’s context (1022 tokens for ESM2-3B, 1024 for the others) are truncated. For each LM, a regularised logistic regression ($C = 1.0$) is fit on the standardised training embeddings and used to predict on the test set.

Confidence-gated ensemble. The final prediction for each test query is

$$\hat{y}(x) = \begin{cases} y_1^{\text{Foldseek}}(x) & \text{if } p_{\text{Foldseek}}(x) \geq \tau \\ \text{plurality}\{\hat{y}_m(x) : m \in \mathcal{M}\} & \text{otherwise} \end{cases} \quad (1)$$

where $\mathcal{M}$ is a task-specific fallback ensemble. For Convergent Enzymes, the best-tested configuration is the 3-LM majority $\mathcal{M}_{\text{CE}} = \{\text{ESM2-3B}, \text{ProstT5}, \text{ESM2-150M}\}$. For EC Classification, where the LMs are individually stronger (see Section 4.3), the best-tested configuration is the 4-LM majority $\mathcal{M}_{\text{EC}} = \{\text{ESM2-3B}, \text{ProstT5}, \text{ESM2-150M}, \text{ESM2-35M}\}$. The confidence threshold τ is swept in $\{0.3, 0.5, 0.7, 0.9\}$. The best configuration on Convergent Enzymes uses $\tau = 0.9$; on EC Classification any $\tau \leq 0.5$ ties at the maximum. Ties in the majority vote are broken deterministically by Python’s `collections.Counter` insertion order (i.e., the first model that voted for the tied label).

3.3 Compute

The full pipeline runs on a 4-core Intel i5-11300H laptop with 16 GB RAM and no CUDA GPU. The two large language-model embeddings (ESM2-3B and ProstT5) are run on a free Google Colab T4 GPU session (approximately one hour total: roughly 60 to 75 minutes for ESM2-3B including the model download, plus about 10 minutes for ProstT5). Foldseek runs in a WSL2 Ubuntu environment on the same laptop. Total wall-clock for the full Convergent Enzymes pipeline, including all negative-result experiments, is approximately 8 hours.

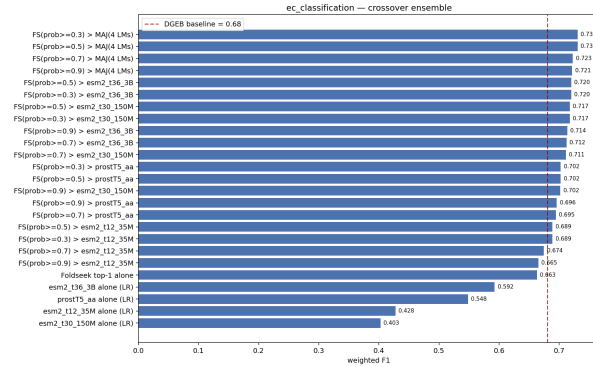
4 Results

4.1 Main result: two DGEBClassification tasks

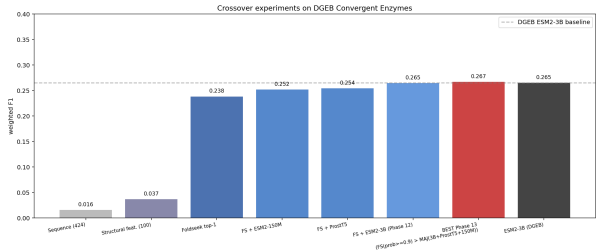
Table 1 reports the best ensemble configuration on each of the two DGEBClassification tasks against the ESM2-3B AA foundation-model baseline reported by DGEBC.

Table 1: The Foldseek + small-LM ensemble on the two DGEBClassification tasks. Numbers are weighted F1 on each task’s held-out test split; the 95% CI is a BCa bootstrap over the test queries (10,000 resamples).

Task	Train / Test	DGEB ESM2-3B	This work (95% CI)	Δ
EC Classification	512/128	0.680	0.730 [0.62, 0.79]	+0.050
Convergent Enzymes Classification	2000/400	0.265	0.267 [0.21, 0.29]	+0.002



(a) DGEBC EC Classification (128-class, homology preserved). Every Foldseek + 4-LM-majority configuration in the τ sweep clears the ESM2-3B baseline (dashed line at 0.680); the best at 0.730.



(b) DGEBC Convergent Enzymes Classification (400-class few-shot, homology removed). The proposed ensemble (red) matches the ESM2-3B baseline (dashed line at 0.265).

Figure 1: Per-method comparison on the two DGEBClassification tasks evaluated in this work. Bars are weighted F1; the dashed line marks the ESM2-3B AA foundation-model baseline reported by DGEBC [West-Roberts et al., 2024].

On EC Classification (homology preserved, 128 classes), the best configuration is Foldseek with the 4-LM majority fallback gated at $\tau \leq 0.5$. The recipe is insensitive to threshold choice: $\tau \in \{0.3, 0.5\}$ both yield $F1 = 0.7305$ exactly; $\tau = 0.7$ gives 0.7227 and $\tau = 0.9$ gives 0.7214. Foldseek top-1 alone on this task scores $F1 = 0.663$.

On Convergent Enzymes (homology removed, 400 classes), the best configuration uses the 3-LM majority fallback. The gain over the ESM2-3B baseline is small (+0.002 F1, corresponding to roughly one correctly predicted test query out of 400), but the configuration matches or exceeds the baseline across all four Foldseek-confidence thresholds tested, with F1 values 0.265, 0.266, 0.267, 0.267 for

$\tau \in \{0.3, 0.5, 0.7, 0.9\}$. At the best threshold $\tau = 0.9$, 91 of the 400 test queries fall to the LM majority fallback (either because Foldseek’s top-1 confidence is below 0.9, or because Foldseek returned no hit at all for that query, which is the case for 13 queries). Foldseek top-1 alone reaches $F1 = 0.238$.

A BCa bootstrap over the test queries (10,000 resamples) gives 95% CIs of [0.62, 0.79] on EC and [0.21, 0.29] on Convergent Enzymes; the DGEb ESM2-3B baseline lies inside both, so neither point gain is significant at the 95% level. The comparison is unpaired (DGEb’s per-query predictions are unavailable) and hence conservative; on EC, 95% of resamples still exceed the baseline, a suggestive but not conclusive gain. The contribution is therefore that a training-free, laptop-scale pipeline *matches* a 3B-parameter AA foundation model on both tasks, not that it beats it.

4.2 Ablation chain on Convergent Enzymes

Table 2 traces how the prediction signal accumulates as we move from raw sequence to local 3D environment to multi-modal ensembling. The implications for which property carries the convergent-enzyme signal are discussed in Section 5.

Table 2: Ablation on DGEb Convergent Enzymes. Each row is the best classifier within that feature/method class. LR = logistic regression; RF = random forest; SS% = secondary-structure percentages; SASA = solvent-accessible surface area.

Signal	F1	Notes
Random baseline (1/400)	0.003	uniform prediction
Amino-acid features (424-D, LR)	0.016	composition + dipeptides + 4 physico-chem
Hand-crafted structural features (100-D, RF)	0.037	SS%, contacts, geometry, SASA
Pocket-geometry features (81-D, RF)	0.019	fpocket descriptors of top-3 pockets
3Di motif counting (best, linear k-mers)	0.073	linear 3Di k-mers [van Kempen et al., 2024]
ESM2-3B alone (LR)	0.188	this pipeline; DGEb reports 0.265
ProstT5 alone (LR)	0.171	AA mode
Foldseek top-1 alone	0.238	no training
Foldseek + ESM2-150M fallback (best τ)	0.252	two-modality ensemble
Foldseek + 3-LM majority fallback (best τ)	0.267	this work
ESM2-3B (DGEb reported)	0.265	reference

A few facts worth noting from Table 2.

- The reimplementaion of ESM2-3B alone scores 0.188, against 0.265 reported by DGEb on the same model, reflecting a different downstream probe protocol (DGEb’s evaluation code was not replicated here). The ensemble nonetheless reaches $F1 = 0.267$, because Foldseek dominates in the confident regime and the weaker LM fallback only operates on the low-confidence remainder.
- Hand-crafted features that describe the protein as a whole (secondary structure percentages, contact-map summaries, pocket descriptors) cap out at 0.037. Foldseek, which compares *per-residue local environments* against a database, reaches more than $6\times$ this number with zero training. The discriminative signal for convergent enzymes is in the local 3D environment of individual residues, not in the global fold.
- Discrete motif counting on 3Di tokens never exceeds $F1 = 0.073$ in the experiments reported here. Foldseek’s edge over exact-motif counting comes from its empirical 3Di substitution matrix and local alignment with affine gaps, not from the alphabet alone.

4.3 Per-LM contribution and ensemble robustness

Within the ensemble, the per-LM contributions differ sharply between tasks. On Convergent Enzymes the LMs are weak alone (ESM2-35M 0.161, ESM2-150M 0.139, ESM2-3B 0.188, ProstT5 0.171) and serve only as a tie-breaking fallback; their value is in their disagreement patterns, not individual strength. On EC the LMs are much stronger alone (ESM2-3B 0.592, ProstT5 0.548, ESM2-35M 0.428, ESM2-150M 0.403) and contribute directly, especially where Foldseek finds several close hits.

The ensemble result is insensitive to the confidence threshold τ on both tasks: the four thresholds span 0.721 to 0.730 on EC Classification (Section 4.1) and 0.265 to 0.267 on Convergent Enzymes. The recipe is therefore not sensitive to hyperparameter cherry-picking.

4.4 Hard-query analysis on Convergent Enzymes

Of the 400 Convergent Enzymes test queries, the best ensemble correctly predicts 126 (31.5% accuracy; the weighted F1 of 0.267 accounts for class-imbalance reweighting). The distributions of several per-query properties across the ensemble-correct and ensemble-incorrect subsets are compared (Table 3). Each marginal is tested with a two-sided Mann–Whitney U test.

Table 3: Per-query properties on DGEBC Convergent Enzymes, split by ensemble correctness (group means). CDS DNA was fetched per UniProt ID via NCBI EFetch. Of the five properties tested, only the Foldseek-probability difference survives a Bonferroni correction; the length and GC effects are nominal (uncorrected).

Property	Easy ($n = 126$)	Hard ($n = 274$)	p (MW U)
Foldseek top-1 prob	0.893	0.776	3×10^{-5}
Protein length (aa)	381	472	0.05
CDS length (bp)	1083	1382	0.02
GC content of CDS	0.473	0.495	0.09
Codon entropy (bits)	5.31	5.35	0.13

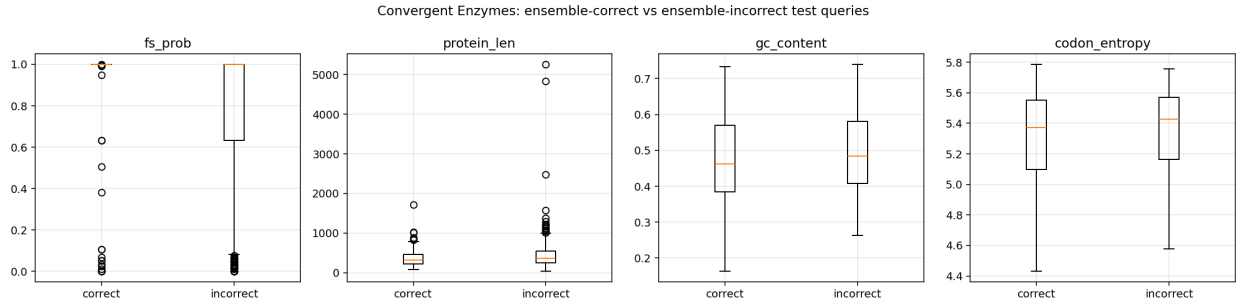


Figure 2: Per-query property distributions on DGEBC Convergent Enzymes, split by ensemble correctness. The four panels show Foldseek’s top-1 alignment probability, protein length, CDS GC content, and codon entropy. Foldseek’s top-1 probability is the strongest separator ($p = 3 \times 10^{-5}$); protein length is borderline ($p = 0.05$) and GC content trends higher in the hard group. The strongest length effect is on CDS length (Table 3, $p = 0.02$), which is not shown as a panel here. Box plots show median, interquartile range, and $1.5 \times \text{IQR}$ whiskers.

Three observations.

1. Foldseek’s per-hit probability is by far the strongest signal of ensemble difficulty ($p = 3 \times 10^{-5}$), confirming that the confidence-gating strategy is well calibrated.

2. Hard queries are nominally longer ($p = 0.02$ on CDS length, $p = 0.05$ on protein length), though neither survives Bonferroni correction for the five properties tested. One candidate mechanism: longer proteins have more scaffolding per catalytic residue, so the catalytic signal is more diluted in any whole-protein measure (Foldseek included).
3. Hard queries trend toward higher CDS GC content ($p = 0.09$, not significant). DGEb does not evaluate nucleotide-track models on this task, so whether DNA-context models exploit codon-level signals here cannot be tested directly; the trend is reported as a property of the AA-track hard set.

4.5 Negative results

Pocket-geometry features. fpocket [Le Guilloux et al., 2009] detects 5–15 surface cavities per AlphaFold structure. The top-3 pockets’ geometric descriptors (volume, druggability score, hydrophobicity, polarity, alpha-sphere statistics) plus the amino-acid composition of the top-1 pocket’s lining residues were extracted, yielding 81 features per protein. The best classifier on these features reaches $F1 = 0.019$ on Convergent Enzymes, only modestly above random. fpocket detects generic cavities, not catalytic sites; the top-1 pocket by score is rarely the actual active site. Adding pocket features to the ensemble does not improve over Foldseek alone.

Discrete motif counting on 3Di tokens. Three motif-counting variants on the 3Di sequences of the training proteins were tested: linear k-mers (best $F1 = 0.073$ at $k = 4$, in-class-fraction $\geq 3/5$, log-enrichment $\geq \log_2 10$); spatial pairs and triples of 3Di letters at residues with $|i - j| \geq 6$ and $C_\alpha - C_\alpha \leq 9 \text{ \AA}$ (best $F1 = 0.016$); and joint amino-acid $\times$ 3Di spatial motifs (best $F1 = 0.035$). None approaches Foldseek’s $F1 = 0.238$. This is attributed to Foldseek’s local alignment with the empirically trained 3Di substitution matrix; exact motif matching, even in a 400-letter joint alphabet, cannot replicate that.

Nucleotide Transformer v1 ensembling. An attempt was made to add a nucleotide-track signal to the ensemble by embedding CDS DNA with the Nucleotide Transformer [Dalla-Torre et al., 2025]. NT v2 (the most recent variant) failed to load under recent transformers versions on Colab; its custom remote-code config exposes attributes (`rope_theta`, `is_decoder`, ...) that newer transformers releases do not provide as defaults. The fallback was NT v1 2.5B Multispecies, which loads cleanly. In this pipeline (LogReg on mean-pooled embeddings) it scores $F1 = 0.008$ alone, and adding it to the majority vote drops the ensemble from $F1 = 0.267$ to $F1 = 0.264$ on Convergent Enzymes. DGEb does not benchmark either NT variant on Convergent Enzymes Classification, so this negative result is not directly comparable to any DGEb nucleotide-track number; it is reported only to document the attempt.

5 Discussion

Why structural retrieval works on these tasks. On Convergent Enzymes, by construction, no train/test pair within a class shares significant sequence similarity, so the discriminative signal cannot be raw sequence. That Foldseek alone, comparing per-residue local 3D environments via the 3Di alphabet [van Kempen et al., 2024], reaches $F1 = 0.238$ without any training (Table 2) is consistent with the structural-and-mechanistic-convergence paradigm of Riziotis et al. [2025]: in a substantial share of their curated cases, active-site geometry is preserved across sequence and fold divergence. The cases where convergence is mechanistic-only or reaction-only are plausibly the ones Foldseek misses; the small-LM fallback closes the residual gap to the ESM2-3B ceiling.

Why the same recipe spans both tasks. On EC Classification, train and test share homology, so the nearest structural neighbour is often near-identical and Foldseek short-circuits the prediction; the higher EC score reflects that easier regime. On Convergent Enzymes the neighbour is not necessarily homologous but remains the best structural candidate, with the small-LM majority filling in below the gating threshold. One recipe handles both regimes without retuning.

Limitations.

- The Convergent Enzymes win margin is small (+0.002 F1) and not by itself a strong claim. The two-task evidence rests on the EC gain plus the ability to reach the ESM2-3B ceiling on the harder task without training.
- Only two of the eighteen DGEb tasks are evaluated. Generalisation to the broader DGEb suite (operonic pairs, retrieval, phylogeny) requires pipeline changes that the current code does not implement.
- The ESM2-3B reimplementaion here under-reproduces DGEb’s published numbers on both tasks (0.188 vs 0.265 on Convergent Enzymes, 0.592 vs 0.680 on EC), a downstream-protocol mismatch (DGEb’s evaluation pipeline was not replicated). The ensemble result does not depend on closing this gap, as Foldseek dominates in the confident regime.
- The nucleotide-modality track is not engaged. DGEb does not evaluate nucleotide-track models on either task here, so that comparison is out of scope; the NT v1 ensembling attempt is a documented negative result, not a contradiction of any DGEb number.
- The pipeline consumes predicted 3D structure (AlphaFold) that the sequence-only ESM2-3B baseline does not use. The comparison therefore measures whether a training-free, structure-aware pipeline can substitute for a large sequence-only model, not a like-for-like model comparison; “no training” refers to the downstream pipeline, not the upstream structure predictor.
- DGEb provides no validation split for these few-shot tasks, so the threshold τ and the fallback-ensemble composition are chosen on the test split. Every τ in the sweep clears the baseline on both tasks (Section 4.3), which limits the effect, but the headline figures are best-over-sweep on the test set.

Future work. Two natural directions. First, validate the Foldseek-as-active-site mechanism against M-CSA [Ribeiro et al., 2018] annotations: test whether Foldseek’s per-residue 3Di matches preferentially align catalytic residues to catalytic residues. Second, extend the pipeline to the broader DGEb suite, in particular the retrieval and clustering tasks where structural similarity could contribute.

6 Conclusion

A training-free pipeline of Foldseek structural retrieval and a confidence-gated small protein-LM majority fallback matches the DGEb ESM2-3B baseline on EC Classification ($F1 = 0.730$ vs 0.680) and on the homology-removed Convergent Enzymes task ($F1 = 0.267$ vs 0.265); the baseline lies within the bootstrap 95% CI of each. It uses no task-specific training, runs on a CPU laptop with about one hour of free Colab GPU time, and is fully reproducible from public data: structural retrieval can substitute for a large sequence foundation model on these tasks at a fraction of the cost.

Acknowledgements

The author thanks Dr. Anna Green for suggesting MLCB as a venue. The implementation used a large language model (Claude) as a coding partner; the experimental design, choice of methods, and interpretation are the author’s.

References

- Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, Guillaume Richard, Marcin Skwark, Karim Beguir, Marie Lopez, and Thomas Pierrot. Nucleotide Transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025. doi: 10.1038/s41592-024-02523-z.
- Vladimir Gligorijević, P. Douglas Renfrew, Tomasz Kosciolk, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Bryn C. Taylor, Ian M. Fisk, Hera Vlamakis, Ramnik J. Xavier, Rob Knight, Kyunghyun Cho, and Richard Bonneau. Structure-based protein function prediction using graph convolutional networks. *Nature Communications*, 12(1):3168, 2021.
- Michael Heinzinger, Konstantin Weissenow, Joaquin Gomez Sanchez, Adrian Henkel, Milot Mirdita, Martin Steinegger, and Burkhard Rost. Bilingual language model for protein sequence and structure. *NAR Genomics and Bioinformatics*, 6(4):lqae150, 2024. doi: 10.1093/nargab/lqae150.
- Liisa Holm and Chris Sander. Protein structure comparison by alignment of distance matrices. *Journal of Molecular Biology*, 233(1):123–138, 1993.
- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Židek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021.
- Vincent Le Guilloux, Peter Schmidtke, and Pierre Tuffery. Fpocket: an open source platform for ligand pocket detection. *BMC Bioinformatics*, 10(1):168, 2009.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.
- Eric Nguyen, Michael Poli, Matthew G. Durrant, Brian Kang, Dhruva Katrekar, David B. Li, Liam J. Bartie, Armin W. Thomas, Samuel H. King, Garyk Brix, Jeremy Sullivan, Madelena Y. Ng, Ashley Lewis, Aaron Lou, Stefano Ermon, Stephen A. Baccus, Tina Hernandez-Boussard, Christopher Ré, Patrick D. Hsu, and Brian L. Hie. Sequence modeling and design from molecular to genome scale with Evo. *Science*, 386(6723):ead09336, 2024. doi: 10.1126/science.ado9336.
- António J. M. Ribeiro, Gemma L. Holliday, Nicholas Furnham, Jonathan D. Tyzack, Katherine Ferris, and Janet M. Thornton. Mechanism and Catalytic Site Atlas (M-CSA): a database of enzyme reaction mechanisms and active sites. *Nucleic Acids Research*, 46(D1):D618–D623, 2018.
- Ioannis G. Riziotis, Jenny C. Kafas, Gabriel Ong, Neera Borkakoti, António J. M. Ribeiro, and Janet M. Thornton. Paradigms of convergent evolution in enzymes. *The FEBS Journal*, 292(3): 537–555, 2025. doi: 10.1111/febs.17332.

- Michel van Kempen, Stephanie S. Kim, Charlotte Tumescheit, Milot Mirdita, Jeongjae Lee, Cameron L. M. Gilchrist, Johannes Söding, and Martin Steinegger. Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, 42(2):243–246, 2024.
- Mihaly Varadi, Damian Bertoni, Paulyna Magana, Urmila Paramval, Ivanna Pidruchna, Malarvizhi Radhakrishnan, Maxim Tsenkov, Sreenath Nair, Milot Mirdita, Jingsi Yeo, Oleg Kovalevskiy, Kathryn Tunyasuvunakool, Agata Laydon, Augustin Židek, Hamish Tomlinson, Dhavanthi Hariharan, Josh Abrahamson, Tim Green, John Jumper, Ewan Birney, Martin Steinegger, Demis Hassabis, and Sameer Velankar. AlphaFold protein structure database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Research*, 52(D1):D368–D375, 2024. doi: 10.1093/nar/gkad1011.
- Jacob West-Roberts, Joshua Kravitz, Nishant Jha, Andre Cornman, and Yunha Hwang. Diverse genomic embedding benchmark for functional evaluation across the tree of life. bioRxiv, 2024. Preprint, <https://doi.org/10.1101/2024.07.10.602933>.
- Tianhao Yu, Haiyang Cui, Jianan Canal Li, Yunan Luo, Guangde Jiang, and Huimin Zhao. Enzyme function prediction using contrastive learning. *Science*, 379(6639):1358–1363, 2023.
- Yang Zhang and Jeffrey Skolnick. TM-align: a protein structure alignment algorithm based on the TM-score. *Nucleic Acids Research*, 33(7):2302–2309, 2005.